

Training and Testing

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Distinguish frames from groups for Splits and related examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is training P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared groups for each pair of sets. Repair it by assigning P and Q to training, R to validation and S to test. List all frames in each role and give row and group counts. State one limit of the repaired plan.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased unseen test result. Explain the issue and revise the evaluation plan.

6. Solve the given case

A rule is developed and evaluated using the same books A–H.

Why does this not establish unseen performance?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test evidence?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 11 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Keep final user profiles hidden until weights and thresholds are fixed.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased unseen test result. Explain the issue and revise the evaluation plan.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the three roles

Explain what training, validation and test sets are used for. What changes if you repeatedly choose a model using its test score?

My reasoning and evidence

13. W2 Distinguish frames from groups

For the worked A through F plan, give training, validation and test counts first in groups and then in rows. Interpret [] from the code and state one problem it cannot detect.

My reasoning and evidence

14. W3 Repair a related example leak

A1 and A2 are in training and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

My reasoning and evidence

15. W4 Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is training P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared groups for each pair of sets. Repair it by assigning P and Q to training, R to validation and S to test. List all frames in each role and give row and group counts. State one limit of the repaired plan.

My reasoning and evidence

16. W5 Match separation to the claim

A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative. Is this enough for a strong performance claim?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Training and Testing

A fictional catalogue model learns from scans of book covers and will face previously unseen books.

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed training A1,B1,C1; validation A2,D1; test B2,C2,D2.

Training and Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Choose unit

The intended kind of new example being evaluated is a new book, not merely a new scan.



2. Find leakage

A crosses train/validation; B,C cross train/test; D crosses validation/test.



3. Group repair

Training all A,B scans; validation all C scans; test all D scans.



4. Freeze assignment

Keep every scan from one book in exactly one role before fitting.

CHANGE ONE THING

Give A2 a new filename without changing the book it depicts.

Explain the changed result using the matching lesson.

How many distinct books are in each repaired role, and why report that as well as scan counts?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue model learns from scans of book covers and will face previously unseen books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep every scan from one book in exactly one role before fitting.

1. Group repair
2. Choose unit
3. Find leakage
4. Freeze assignment

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Freeze assignment Keep every scan from one book in exactly one role before fitting. Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. Grouping by book matches the intended claim about new books and makes the split inspectable. The repaired split has four training, two validation and two test scans, but only two, one and one books respectively.

Practice 2: explain the deciding evidence

Changed-case practice: Give A2 a new filename without changing the book it depicts. Which conclusion is supported by the supplied facts?

1. Its group is still A; the original related examples still cross the split.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Its group is still A; the original related examples still cross the split. Renaming a row does not remove the relationship between scans.

Repair a missing building block

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated input format, selected rule and recorded output. Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?. Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
